

The effect of radial and non-radial stellar oscillations on the light, colour and velocity variations

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Summary. The effect of radial and non-radial stellar oscillations on the observed light, colour and velocity variations is examined. A method of estimating the phase lag, radius and other pulsation constants is applied to classical cepheids. Amplitude and phase relations for non-radially oscillating stars are discussed with particular emphasis on the quadrupole mode. The radial pulsation and quadrupole oscillation models for β Cep stars are discussed. Some effects are predicted for multiperiodic stars and a method of determining the radius and pulsation modes is suggested.

1 Introduction

The identification of non-radial oscillations is one of the most basic, but one of the most difficult problems in observations of variable stars. The method most often used is to compare the observed frequency spectrum with one calculated from a model. This is sensitive to the physics of the model.

Dziembowski (1977) has suggested a model-independent method of determining the oscillation mode by application of the Wesselink technique. He showed that the resulting Wesselink radius is sensitive to the spherical harmonic order of the oscillation. Balona & Stobie (1979) have extended Dziembowski's ideas and formulated the practical application of the Wesselink technique for this purpose.

In this note we discuss further implications of radial and non-radial oscillations for the observed light, colour and velocity variations. We propose a simple method to determine the phase lag and other parameters for radial oscillations and apply it to classical cepheids. We also examine the phase relationships between the light, colour and velocity curves for non-radially oscillating stars, apply the method to β Cep variables and predict some results for multiperiodic stars.

2 Radial pulsation

As discussed by Balona & Stobie (1979), the concept of effective temperature is not a useful one in dealing with oscillating stars. Instead we use the surface brightness $S = -2.5 \log F$,

where $F = L/A$ is the observed luminosity per unit area. Wesselink (1969) has shown that the surface brightness in the V band is strongly correlated with a colour index such as $B-V$ and is an observable quantity.

For a star in radial pulsation we can express the radius dependence as:

$$R = R_0(1 + \epsilon \cos \omega t)$$

and the flux dependence at any point on the photosphere as:

$$F = F_0[1 + \epsilon f \cos(\omega t + \psi)].$$

The quantity f determines the relative importance of the flux variation in the total luminosity variation. If the luminosity in the V band is measured, rather than the bolometric luminosity, the corresponding value of f should be denoted by f_v . This is usually clear from the context and the subscript will not be used in the remainder of this paper. The phase lag, ψ , as we have defined it, is not directly related to the phase difference between the light and velocity variations, but is of a more fundamental nature.

If we assume that the pulsation amplitude $\epsilon \ll 1$ and apply the linear approximation, it can be shown on integration over the stellar surface that:

$$\Delta L/L = \epsilon(f^2 + 4f \cos \psi + 4)^{1/2} \cos(\omega t + \alpha) \quad (1)$$

$$\Delta F/F = \epsilon f \cos(\omega t + \psi) \quad (2)$$

$$\Delta V_r = 0.708 \epsilon \omega R_0 \cos(\omega t - \pi/2) \quad (3)$$

$$\tan \alpha = f \sin \psi / (2 + f \cos \psi) \quad (4)$$

where $\Delta L/L$ is the brightness variation, $\Delta F/F$ the flux variation (which is proportional to the colour variation) and ΔV_r is the velocity variation relative to the observer.

If the light, colour and velocity observations are decomposed into sinusoidal components, it can be shown that this system of equations holds for each harmonic component provided the amplitude is not too large. It is then possible to solve for ψ , f , ϵ and R_0 even though the variations may be far from sinusoidal. The procedure is simple: ψ is obtained from the phasing of the colour and velocities, α from the light and velocities, f from equation (4), ϵ from the light amplitude and R_0 from the velocity amplitude.

3 Application to cepheids

To determine the extent to which the above relations are valid in practice, we applied the method to a Fourier analysis of about 60 well observed cepheids. Since R_0 depends on the previously determined values of ψ , f and ϵ , it follows that if we can show that the radius is reasonably well determined then the other quantities will also be at least as well determined. That this is in fact the case can be seen from Fig. 1 which shows the resulting period–radius relationship. Comparison with the more refined maximum likelihood radii of Balona (1977) shows very good agreement except for some cepheids of low amplitude. As expected, the scatter in Fig. 1 is larger than the corresponding diagram of Balona (1977), but is still small enough to provide confidence in the values of ψ , f and ϵ . These values are given in Table 1.

Adiabatic pulsation theory predicts that the phase lag, ψ , should be 180° for all pulsating stars. The fact that it is much closer to 90° for stars in the cepheid instability strip was not completely understood until Castor (1968) showed that it was mainly due to the hydrogen ionization zone. For cepheids the phase lag appears to be randomly distributed about the mean value $\langle \psi \rangle = 124^\circ.3$ with a rms scatter of $12^\circ.4$ and is independent of period or effective temperature.

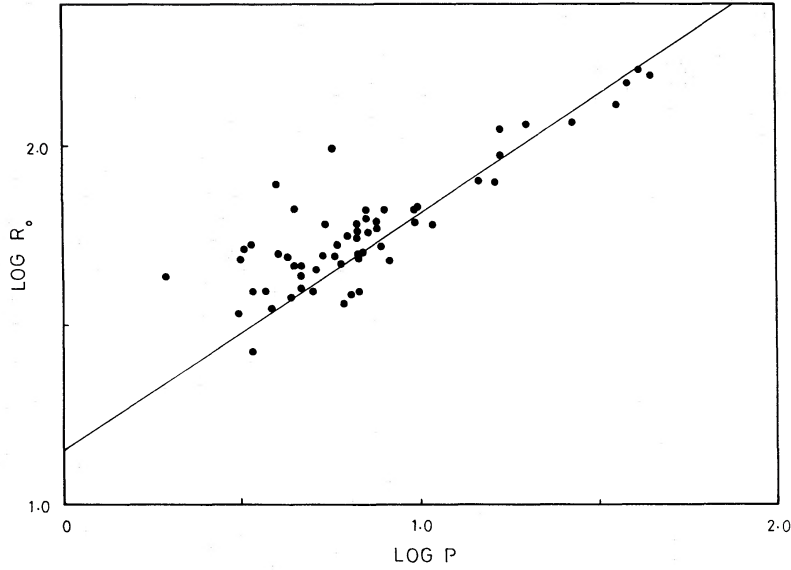


Figure 1. The period–radius relationship for classical cepheids as derived from the sinusoidal components of the V , $B-V$ and velocity curves. The straight line is derived from the Sandage & Tammann (1969) PLC relationship.

In a study of RR Lyrae models, Christy (1966) showed that his quantity $\mathcal{R}$ ($= 2\pi(f^2 + 4f \cos \psi + 4)^{-1/2}$ in our notation) which is quite a useful probe, varies strongly with effective temperature. He finds values of $f = 2.4$ at the low temperature boundary and $f = 7.6$ at high temperatures. We note also that Dziembowski (1977) finds a similar trend for δ Scuti stars. We examined the cepheid models of Stobie (1969a, b, c) but could not find such a trend. The observed trend for cepheids is shown in Fig. 2.

Finally, it is easy to see that equations (1) and (3) give a relation between the velocity and light amplitudes:

$$A_{RV}/A_V = 4.097 R_0 / [P(f^2 + 4f \cos \psi + 4)^{1/2}].$$

Even for constant f and ψ this ratio depends on R_0/P which is not a constant. Fig. 3 shows the A_{RV} versus A_V diagram using the amplitudes of the fundamental and first harmonic (i.e. $\cos 2\omega t$ term) components of the variation. It is evident that deviation from a straight line occurs at larger amplitudes for which the period is the longest.

4 Non-radial oscillations

Equations (1)–(4) can be generalized to the case of pure non-radial oscillation, i.e. where the displacement and flux variation can be described by a spherical harmonic of specific order. This has been discussed by Balona & Stobie (1979). The result is:

$$\Delta L/L = \epsilon P_{lm}(\mu_0) \{f^2 b_l^2 + g_l^2 + 2fb_l g_l \cos \psi\}^{1/2} \cos(\omega t + m\phi_0 + \alpha) \quad (5)$$

$$\Delta F/F = \epsilon P_{lm}(\mu_0) \{f^2 b_l^2 + h_l^2 + 2fb_l h_l \cos \psi\}^{1/2} \cos(\omega t + m\phi_0 + \beta) \quad (6)$$

$$\Delta V_r = \epsilon P_{lm}(\mu_0) \omega R_0 (u_l + \alpha_H v_l) \cos(\omega t + m\phi_0 - \pi/2) \quad (7)$$

$$\tan \alpha = fb_l \sin \psi / (g_l + fb_l \cos \psi) \quad (8)$$

$$\tan \beta = fb_l \sin \psi / (h_l + fb_l \cos \psi) \quad (9)$$

Table 1. Pulsation properties of classical cepheids determined from the fundamental sinusoidal components of the V , $B-V$ and radial velocity variations. The phase lag, ψ , is in degrees; f_V is the ratio of flux variation in the V band to the radius variation. The semi-amplitude of the radius variation, ϵ , is relative to the mean radius R_0 in solar radii. A is the slope of the surface brightness versus $B-V$ relationship.

Star	Period	ψ	f_V	ϵ	R_0	A
U Aql	7.023930	112.6	6.1	0.051	63.2	2.041
FF Aql	4.470959	139.8	11.8	0.014	67.4	2.399
V496 Aql	6.806900	115.1	5.5	0.031	39.8	2.053
η Aql	7.176641	118.5	6.6	0.052	59.4	2.057
RT Aur	3.728261	142.9	7.0	0.056	27.0	2.478
U Car	38.756000	132.9	4.0	0.136	152.5	2.204
V Car	6.696380	122.7	5.8	0.046	51.1	2.030
UX Car	3.682246	117.6	9.5	0.037	41.3	2.307
ER Car	7.718700	137.1	5.5	0.053	52.8	2.284
λ Car	35.535100	136.4	3.5	0.128	131.9	2.220
SU Cas	1.949298	110.0	15.6	0.012	43.6	2.401
V Cen	5.493920	119.4	8.7	0.039	61.4	2.168
XX Cen	10.953700	125.9	5.3	0.079	60.7	2.096
AZ Cen	3.212055	113.7	13.8	0.012	52.3	2.205
BB Cen	3.997660	134.0	17.0	0.012	80.4	2.206
δ Cep	5.366341	124.4	7.9	0.047	49.6	2.067
RY CMa	4.678450	123.4	6.7	0.048	40.6	2.293
VZ CMa	3.126260	114.7	11.6	0.017	50.4	2.312
R Cru	5.825650	115.4	7.2	0.045	54.0	2.055
S Cru	4.689820	119.1	7.6	0.041	47.4	2.133
T Cru	6.733190	125.2	5.7	0.040	59.8	2.053
AG Cru	3.837280	121.9	8.4	0.044	36.0	2.329
BG Cru	3.342700	127.0	13.6	0.008	54.7	2.365
X Cyg	16.386600	146.0	3.8	0.153	80.0	2.211
TX Cyg	14.708250	131.6	4.7	0.112	80.7	2.373
CD Cyg	17.071380	127.3	4.7	0.114	95.8	1.988
TX Del	6.165800	152.3	5.0	0.078	36.6	2.153
β Dor	9.842336	125.9	5.4	0.055	68.9	1.973
T Mon	27.020500	136.7	3.9	0.144	118.8	2.452
R Mus	7.510500	123.1	7.6	0.049	59.6	2.107
S Mus	9.660200	124.3	5.2	0.053	61.8	2.603
RT Mus	3.086170	112.2	8.2	0.040	34.7	2.264
S Nor	9.754160	98.3	4.9	0.052	67.3	1.714
Y Oph	17.1220	112.7	6.3	0.036	113.7	2.008
BF Oph	4.067750	120.9	7.5	0.039	50.3	2.133
AW Per	6.463420	123.2	5.3	0.070	38.9	2.828
RS Pup	41.387600	139.8	3.8	0.163	164.8	2.633
AP Pup	5.084310	126.0	6.4	0.046	45.9	2.154
MY Pup	5.694620	133.7	10.7	0.009	92.0	2.322
RV Sco	6.061330	120.8	7.3	0.048	47.6	2.245
V636 Sco	6.796630	130.2	6.1	0.042	50.0	2.215
S Sge	8.382173	128.1	6.9	0.053	60.8	2.094
U Sgr	6.745301	119.2	6.9	0.045	61.9	2.062
W Sgr	7.594854	125.8	6.9	0.053	62.4	2.112
X Sgr	7.012250	73.8	6.9	0.034	67.5	1.840
Y Sgr	5.773450	136.1	6.7	0.052	50.3	2.356
AP Sgr	5.057930	106.3	7.3	0.043	39.6	2.008
BB Sgr	6.636990	127.4	6.4	0.043	58.4	2.200
R TrA	3.389287	119.5	8.1	0.032	40.1	2.199
S TrA	6.323440	119.5	7.2	0.045	58.1	2.054
T Vel	4.639820	130.5	6.7	0.043	44.2	2.294
V Vel	4.370191	123.6	6.1	0.039	38.2	2.185
RZ Vel	20.398000	140.0	4.5	0.143	117.0	2.352
AH Vel	4.227180	128.4	9.4	0.018	50.6	2.358
BG Vel	6.923570	126.3	5.6	0.040	52.6	2.175
T Vul	4.435572	117.3	7.3	0.039	48.1	2.083
U Vul	7.990676	125.5	7.7	0.043	67.2	2.195
SV Vul	45.100000	137.4	3.3	0.170	158.7	2.198

where we have written

$$g_l = 2b_l - c_l, \quad h_l = 2b_l - c_l - 2(2p_l - q_l), \quad \alpha_H = G\mathcal{M}/\omega^2 R_0^3$$

(G is the gravitational constant and $\mathcal{M}$ the mass of the star), $\mu_0 = \cos \theta_0$ where θ_0 is the aspect angle and $P_{lm}(\mu)$ is the Legendre function. The remaining constants are definite

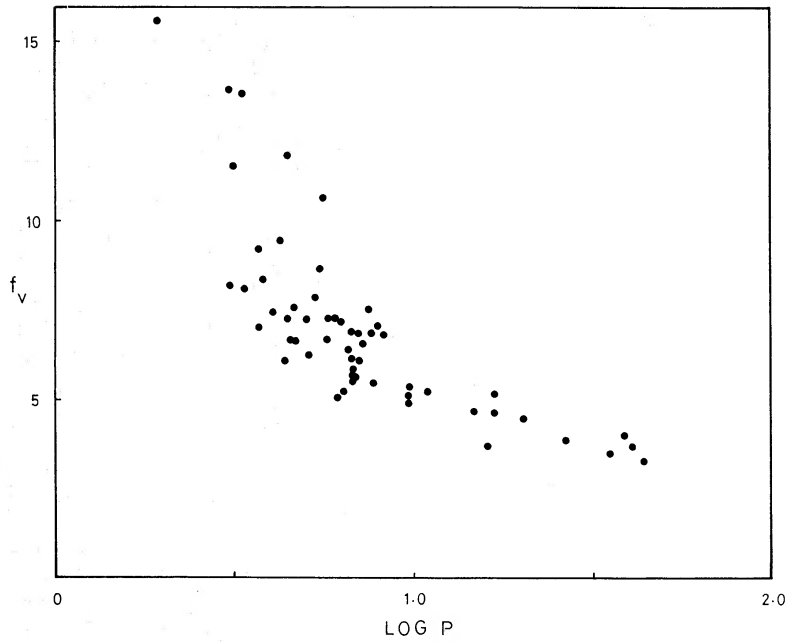


Figure 2. The ratio of flux to radius variation, f , as a function of period.

integrals:

$$u_l = \int_0^1 P_l h \mu^2 d\mu$$

$$v_l = l \int_0^1 h \mu (P_{l-1} - \mu P_l) d\mu$$

$$b_l = \int_0^1 h \mu P_l d\mu$$

$$c_l = l \int_0^1 \left(h + \mu \frac{dh}{d\mu} \right) (P_{l-1} - \mu P_l) d\mu$$

$$p_l = \int_0^1 \mu P_l d\mu$$

$$q_l = l \int_0^1 (P_{l-1} - \mu P_l) d\mu$$

where $\mu = \cos \theta$ ($\theta = 0$ corresponds to the direction of the observer), $P_l = P_l(\mu)$ is the Legendre polynomial and $h = h(\mu)$ is the limb darkening law. When numerical values are given it is assumed that the Eddington limb darkening law $h = 1 + 1.5\mu$ applies and $\alpha_H = 0.08$.

In principle it would be possible to apply these relationships to the amplitudes and phases of the harmonic components to obtain a self-consistent solution for all unknowns, including the spherical harmonic order, l . Note, however, that m cannot be determined from these equations and recourse to line broadening observations or frequency splitting is necessary to

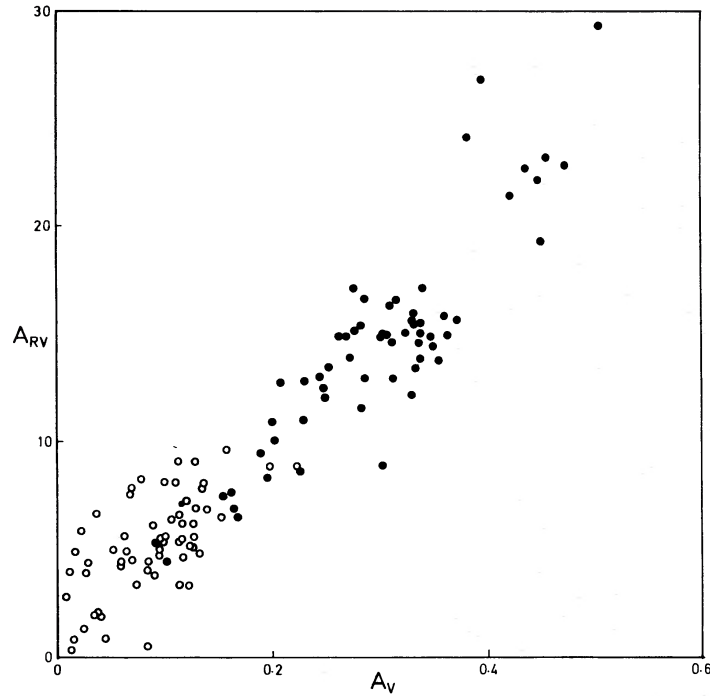


Figure 3. The velocity versus light amplitudes of the fundamental (closed circles) and first harmonic (open circles) sinusoidal components.

identify m . For all odd values of l these equations reduce to the special case $h_l = g_l$, $\alpha = \beta$, $\Delta L/L = \Delta F/F$. Thus the luminosity variations are directly determined by the flux variations. As pointed out by Balona & Stobie (1979), this occurs because for odd values of l the projected area variation is zero and the Wesselink technique breaks down.

In practice there are many difficulties, not the least of which is that those stars for which non-radial oscillation is probable, are likely to be of low amplitude. Accordingly great demands would be required of the accuracy of the observations. Another problem is that most of these stars would probably also be multiperiodic. This is discussed later.

In spite of the problems, it is worthwhile examining some special cases.

(i) *General quadrupole oscillation ($l = 2$)*

The above relationships reduce to

$$\Delta L/L = \epsilon P_{lm}(\mu_0) A \cos(\omega t + m\phi_0 + \alpha) \quad (10)$$

$$\Delta F/F = \epsilon P_{lm}(\mu_0) B \cos(\omega t + m\phi_0 + \beta) \quad (11)$$

$$\Delta V_r = 0.383 \epsilon \omega R_0 P_{lm}(\mu_0) \cos(\omega t + m\phi_0 - \pi/2) \quad (12)$$

$$\tan \alpha = -f \sin \psi / (4 - f \cos \psi) \quad (13)$$

$$\tan \beta = -f \sin \psi / (0.923 - f \cos \psi) \quad (14)$$

where we have written

$$A = 0.325(f^2 - 8f \cos \psi + 16)^{1/2} \quad (15)$$

$$B = 0.325(f^2 - 1.846f \cos \psi + 0.852)^{1/2}. \quad (16)$$

(ii) *Quadrupole oscillation* ($l = 2$), $\psi = 90^\circ$

This case corresponds to stars in the cepheid instability strip and may explain the frequency spectrum of some δ Scuti stars (Shobbrook & Stobie 1974). Equations (13) and (14) become $\tan \alpha = -0.25f$ and $\tan \beta = -1.083f$. From this we immediately deduce the following powerful result:

For quadrupole oscillation of stars in the cepheid instability strip, the phasing of the light and colour variations is in the opposite sense to that of radial pulsators.

Even for large f the phase difference should be easy to detect: for example, when $f = 20$ the difference is 0.024 periods. The phase difference varies rather slowly with ψ , in this example it is 0.019 periods if $\psi = 120^\circ$.

Other non-radial modes with even values of l also give rise to a phase shift $\alpha - \beta$ opposite to that of radial pulsators. However, the magnitude of the phase shift is very much larger than in the quadrupole mode. For example, with $f = 10$, $\psi = 90^\circ$ we find $\alpha - \beta = -0.031$ for $l = 0$, $+0.046$ for $l = 2$, $+0.339$ for $l = 4$, $+0.422$ for $l = 6$ and $+0.455$ for $l = 8$. Thus in principle it should be easy to distinguish between radial, quadrupole and higher order even spherical harmonic orders just from the phasing of the light and colour curve provided that ψ is close to 90° .

(iii) *Quadrupole oscillation* ($l = 2$), $\psi = 180^\circ$

This is the case which might be applicable to β Cephei stars (Osaki 1971). Equations (13) and (14) show that there should be no phase difference between the light and colour variations (as observed), but this also applies to radial pulsation and is only a consequence of the 180° phase lag.

The velocity to light amplitude ratio is

$$A_{RV}/A_V = 6.820(R_0/P)/(f + 4). \quad (17)$$

From the effective temperature and bolometric correction calibrations for early B stars (Heintze 1973), it is easy to show that $\Delta F/F = 2.24 A_{U-B}$. Hence the light to colour amplitude ratio is

$$A_V/A_{U-B} = 2.431(f + 4)/(f + 0.923). \quad (18)$$

Clearly, if we know the radius, equation (17) can be used to estimate f which in turn can be used in equation (18) to predict the light to colour amplitude ratio. Comparison with the observed ratio can be used to test the quadrupole oscillation model.

Similarly for radial pulsation we obtain:

$$A_{RV}/A_V = 4.097(R_0/P)/(f - 2) \quad (19)$$

$$A_V/A_{U-B} = 2.431 - 4.862/f. \quad (20)$$

Table 2 is a list of well-observed β Cep stars taken from Stamford & Watson (1977); the ratio R_0/P was calculated from table II of Balona & Feast (1975). The value of f determined from the velocity to light amplitude ratio and the predicted light to colour amplitude ratio is shown for the radial and quadrupole modes. Although there are many uncertainties, it appears that for some stars radial pulsation is the more likely while for one or two stars quadrupole oscillation produces better agreement. It should be pointed out that for most stars the observations are very uncertain, particularly the $U-B$ amplitudes. For many β Cep

Table 2. Predicted V to $U-B$ amplitude ratios for β Cephei stars assuming radial and quadrupole oscillation models.

Star	Period (days)	A_{RV}/A_V	R_O/P	f	Radial A_V/A_{U-B}	f	Quadrupole A_V/A_{U-B}	Observed A_V/A_{U-B}
γ Peg	0.1518	538	362	4.8	1.4	0.6	7.3	1.2
δ Cet	0.1611	500	350	4.9	1.4	0.8	6.8	1.3
ν Eri	0.1735	414	385	5.8	1.6	2.3	4.8	1.6
	0.1779	425	385	5.7	1.6	2.2	4.8	3.6
β CMa	0.2513	357	328	5.8	1.6	2.3	4.8	2.1
	0.2500	2750	328	2.5	0.5	-3.2	-0.9	
ξ^1 CMa	0.2096	944	460	4.0	1.2	-0.7	36.0	1.3
15 CMa	0.1846	406	439	6.4	1.7	3.4	4.2	4.0
β Cru	0.1605	263	304	6.7	1.7	3.9	4.0	7.9
α Vir	0.1738	552	389	4.9	1.4	0.8	6.8	4.7
τ^1 Lup	0.1774	350	457	7.3	1.8	4.9	3.7	3.3
σ Sco	0.2468	2750	286	2.4	0.4	-3.3	-0.7	1.3
	0.2395	714	286	3.6	1.1	-1.3	-17.4	21.0
λ Sco	0.2137	800	332	3.7	1.1	-1.2	-24.6	2.5
BW Vul	0.2010	1368	404	3.2	0.9	-2.0	-4.5	1.3
12 Lac	0.1931	481	324	4.8	1.4	0.6	7.3	4.6
	0.1974	603	324	4.2	1.3	-0.3	14.4	

stars these data are unavailable. Accurate simultaneous velocity and photometry observations are required for more definite mode identifications.

5 Doubly periodic stars

For stars oscillating with angular frequencies ω_1 , and ω_2 , the general case can be written as:

$$\Delta L/L = A \cos(\omega_1 t + \alpha)$$

$$\Delta F/F = B \cos(\omega_1 t + \beta)$$

$$\Delta V_r = R_0 C \cos(\omega_1 t + \gamma)$$

where

$$A^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos\{(\omega_2 - \omega_1)t + m_2 \phi_2 - m_1 \phi_1 + \alpha_2 - \alpha_1\}$$

a_1 , α_1 being the amplitude and phase of the first oscillation and a_2 , α_2 that of the second as given in equation (5). Similar expressions apply for B and C . In the same way we can write:

$$\tan(\alpha - m_1 \phi_1 - \alpha_1) = \frac{a_2 \sin\{(\omega_2 - \omega_1)t + m_2 \phi_2 - m_1 \phi_1 + \alpha_2 - \alpha_1\}}{a_1 + a_2 \cos\{(\omega_2 - \omega_1)t + m_2 \phi_2 - m_1 \phi_1 + \alpha_2 - \alpha_1\}}$$

with similar expressions for $\tan \beta$ and $\tan \gamma$.

These relationships are intractable as they stand, but it is possible to deduce the following conclusions:

- (i) In general the ratio of light to colour amplitude A/B and velocity to light amplitude

C/A will vary with the beat phase. This is true even if the oscillations belong to spherical harmonics of the same order unless $f_1 = f_2$ and $\psi_1 = \psi_2$.

(ii) The phase difference between the light and colour or light and velocity variations will also vary with the beat phase unless the same conditions apply.

(iii) The relative amplitudes of the two oscillations ϵ_1/ϵ_2 will not be the same as the relative amplitudes of the light, colour or velocity variations:

$$\frac{\Delta L_1}{\Delta L_2} = \left\{ \frac{f_1^2 b_1^2 + g_1^2 + 2f_1 b_1 g_1 \cos \psi_1}{f_2^2 b_2^2 + g_2^2 + 2f_2 b_2 g_2 \cos \psi_2} \right\}^{1/2} \frac{\epsilon_1 P_1}{\epsilon_2 P_2}$$

$$\frac{\Delta F_1}{\Delta F_2} = \left\{ \frac{f_1^2 b_1^2 + h_1^2 + 2f_1 b_1 h_1 \cos \psi_1}{f_2^2 b_2^2 + h_2^2 + 2f_2 b_2 h_2 \cos \psi_2} \right\}^{1/2} \frac{\epsilon_1 P_1}{\epsilon_2 P_2}$$

$$\frac{\Delta V_{r1}}{\Delta V_{r2}} = \left\{ \frac{u_1 + \alpha_H v_1}{u_2 + \alpha_H v_2} \right\} \frac{\omega_1 \epsilon_1 P_1}{\omega_2 \epsilon_2 P_2}$$

where the subscripts refer to the two oscillations as well as the possibly different spherical harmonic orders l and $P = P_{lm}(\mu_0)$.

(iv) The Wesselink method applied to only one or two cycles of ω_1 or ω_2 will give a variable 'radius' if the two oscillations are different in the spherical harmonic order l . From Balona & Stobie (1979) the projected area variation of a single mode may be expressed as

$$\Delta A/A = \beta_l \int V_r dt/R_0.$$

For a doubly periodic variable its projected area variation is

$$\Delta A/A = \beta_1 \int V_{r1} dt/R_0 + \beta_2 \int V_{r2} dt/R_0.$$

This, however, cannot be expressed as $\beta \int (V_{r1} + V_{r2}) dt/R_0$ unless $\beta_1 = \beta_2$. In other words, the projected area variation is no longer proportional to the radial velocity variation unless the modes are of the same spherical harmonic order.

For multiple periodic stars where non-radial modes are suspected, the simplest procedure for determining the radius and the spherical harmonic order is to apply equations (5)–(9) to each sinusoidal component. Considerable simplification will result if some idea of the mode is suspected or if the radius is approximately known. A more rigorous approach would be to prewhiten the data by removing all components except the one under investigation. The results of Balona & Stobie (1979) can then be applied.

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